

Bluestock Mutual Fund Analytics

Final Project Report

End-to-End Data Analytics & Investment Intelligence Capstone

1. Executive Summary

This project develops an end-to-end mutual fund analytics solution covering data ingestion, cleaning, SQLite storage, exploratory analysis, performance analytics, advanced risk and investor analytics, and an interactive Power BI dashboard. The completed Day 6 layer includes historical VaR/CVaR for 40 schemes, rolling 90-day Sharpe analysis for five selected funds, investor cohorts, SIP continuity, a risk-based recommender and sector HHI.

2. Problem Statement

Mutual fund datasets contain scheme, NAV, investor transaction, performance and portfolio information. The objective is to transform these datasets into reliable analytical outputs that support fund comparison, risk assessment, investor behaviour analysis and business decision-making.

3. Objectives

Build a reproducible data workflow; store and query cleaned data in SQLite; perform EDA with 15+ charts; calculate performance and risk metrics; implement advanced analytics; build an interactive four-page Power BI dashboard; and produce professional final documentation.

4. Data & Technology

The project uses mutual fund scheme master, historical NAV, investor transaction, performance and portfolio-holding data. Core technologies are Python, Pandas, NumPy, Matplotlib, Jupyter, SQLite, SQL and Microsoft Power BI.

5. ETL Pipeline

The workflow follows Extract → Transform → Load. Raw data is ingested, standardized and cleaned before analytical use. Important quality controls include date parsing, duplicate handling, missing-value treatment, NAV validation, transaction validation and non-trading-day handling.

6. SQLite & SQL

Cleaned datasets are organized in SQLite to provide a consistent storage and query layer for scheme, NAV, investor, transaction and portfolio information.

7. Exploratory Data Analysis

The EDA stage contains more than 15 visual analyses covering distributions, trends, fund characteristics, transactions and relationships. These analyses provide the evidence base for performance and dashboard insights.

8. Performance Analytics

The performance stage uses daily-return based calculations and risk-adjusted measures, including Sharpe ratio and other project-specified performance measures. The completed notebook and outputs are maintained in the repository.

9. Advanced Analytics

Historical VaR (95%) is the 5th percentile of daily returns and CVaR is the mean of returns at or below that threshold. Rolling 90-day Sharpe uses rolling mean / rolling standard deviation $\times \sqrt{252}$. Investor cohorts are grouped by first transaction year. Investors with 6+ SIPs are assessed using average date gaps, with gaps above 35 days flagged at-risk. The recommender returns the top three funds by Sharpe within Low, Moderate or High risk grade. Sector HHI is $\Sigma(\text{weight}^2)$, with higher values indicating greater concentration.

10. Advanced Findings

The completed SIP continuity analysis identified 1,362 investors with at least six SIP transactions; 1,332 were classified as at-risk under the specified >35-day average-gap rule, giving a 2.2% continuity rate. VaR/CVaR were calculated across 40 schemes, rolling Sharpe was produced for five funds, cohorts were analysed by first transaction year, and HHI was calculated for equity funds.

11. Power BI Dashboard

The dashboard contains four pages: Industry Overview; Fund Performance; Investor Analytics; and SIP & Market Trends. The pages use KPIs, trends, fund comparisons, investor analysis, market comparisons, slicers and the Bluestock visual theme.

12. Business Use Cases

The solution can support risk-adjusted fund comparison, downside-risk monitoring, investor engagement through SIP continuity signals, cohort segmentation, concentration monitoring and risk-profile-based fund screening.

13. Recommendations

Use risk-adjusted metrics alongside raw returns; prioritise investor-engagement efforts using SIP continuity signals; monitor concentrated portfolios with HHI; combine cohort behaviour with fund preference; and treat the simple recommender as a screening tool rather than financial advice.

14. Limitations

The analysis is based on the supplied project datasets and available history. Historical VaR, Sharpe and HHI describe observed data and do not guarantee future performance. The recommender uses risk grade and Sharpe only and is not a complete investment-advice system.

15. Deliverables

The project includes the ingestion/cleaning scripts, SQLite and SQL layer, EDA notebook, performance notebook, Power BI dashboard work, and Day 6 advanced analytics. Day 6 includes `Advanced_Analytics.ipynb`, `recommender.py`, `var_cvar_report.csv`, `rolling_sharpe_chart.png`, `cohort_analysis.csv`, `sip_continuity.csv` and `sector_hhi.csv`.

16. Conclusion

The Bluestock Mutual Fund Analytics project demonstrates a complete analytics workflow from financial data to database storage, statistical analysis, advanced risk and investor analytics, and interactive visualization. It provides a practical foundation for comparing funds, understanding investor behaviour and communicating findings to decision-makers.